

A rule-3 disproof of conjecture 00000000938:  
the Banach–Mazur distance  $d(\ell_1^n, \ell_\infty^n)$  is not  $\sqrt{n}$

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### Abstract

Conjecture 00000000938 asserts, among other things, that the Banach–Mazur distance between  $\ell_1^n$  and  $\ell_\infty^n$  “is exactly  $\sqrt{n}$  (the Goldstine ratio)”. We refute this statement as written: already at  $n = 2$  the linear map  $T(x_1, x_2) = (x_1 + x_2, x_1 - x_2)$  is a bijective isometry from  $\ell_1^2$  onto  $\ell_\infty^2$ , so  $d(\ell_1^2, \ell_\infty^2) = 1 \neq \sqrt{2} \approx 1.41421$ . Moreover,  $\sqrt{n}$  is not merely off by a constant:  $\sqrt{n}$  is the correct value of  $d(\ell_1^n, \ell_2^n) = d(\ell_2^n, \ell_\infty^n)$ , whereas  $d(\ell_1^n, \ell_\infty^n)$  is of order  $n$ . The claim therefore fails at  $n = 2$  and is the wrong quantity in general.

## 1 Definition

**Definition 1** (Banach–Mazur distance). Let  $E, F$  be isomorphic  $n$ -dimensional real Banach spaces. Their *Banach–Mazur distance* is

$$d(E, F) = \inf \{ \|T\| \cdot \|T^{-1}\| : T : E \rightarrow F \text{ an isomorphism} \},$$

where  $\|T\| = \sup_{\|x\|_E \leq 1} \|Tx\|_F$  is the operator norm. Equivalently,  $d(E, F)$  is the infimum of  $\lambda \geq 1$  such that there is an isomorphism  $T$  with  $\lambda^{-1} \|x\|_E \leq \|Tx\|_F \leq \lambda \|x\|_E$  for all  $x \in E$ . In particular  $d(E, F) \geq 1$ , with equality if and only if  $E$  and  $F$  are isometrically isomorphic.

For  $1 \leq p \leq \infty$  we write  $\ell_p^n = (\mathbb{R}^n, \|\cdot\|_p)$ , where  $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$  for  $p < \infty$  and  $\|x\|_\infty = \max_i |x_i|$ . Thus the unit ball of  $\ell_1^2$  is the diamond with vertices  $(\pm 1, 0), (0, \pm 1)$ , while the unit ball of  $\ell_\infty^2$  is the square with corners  $(\pm 1, \pm 1)$ .

The claim under test is the following sentence from conjecture 00000000938:

“the distance between  $\ell_1^n$  and  $\ell_\infty^n$  is exactly  $\sqrt{n}$  (the Goldstine ratio)”.

The text places no restriction on  $n$  and gives no asymptotic qualifier, so the assertion is to be read as an identity valid for all  $n$ . We show it is false already for  $n = 2$ .

## 2 The key isometry identity

**Lemma 2** (Isometry identity). *For all real  $a, b$ ,*

$$\max(|a+b|, |a-b|) = |a| + |b|.$$

*Proof.* By the triangle inequality,  $|a+b| \leq |a|+|b|$  and  $|a-b| \leq |a|+|b|$ , hence  $\max(|a+b|, |a-b|) \leq |a|+|b|$ . For the reverse inequality, split according to the relative signs of  $a$  and  $b$ .

- If  $a \geq 0$  and  $b \geq 0$ , then  $|a+b| = a+b = |a|+|b|$ .
- If  $a \leq 0$  and  $b \leq 0$ , then  $a+b \leq 0$  and  $|a+b| = -(a+b) = |a|+|b|$ .
- If  $a \geq 0 \geq b$ , then  $a-b \geq 0$  and  $|a-b| = a-b = |a|+|b|$ .
- If  $a \leq 0 \leq b$ , then  $a-b \leq 0$  and  $|a-b| = -(a-b) = |a|+|b|$ .

In each case one of  $|a+b|, |a-b|$  equals  $|a|+|b|$ , so  $\max(|a+b|, |a-b|) \geq |a|+|b|$ . This proves the identity.  $\square$

**Proposition 3.** *Define  $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  by  $T(x_1, x_2) = (x_1 + x_2, x_1 - x_2)$ . Then*

$$\|Tx\|_\infty = \max(|x_1 + x_2|, |x_1 - x_2|) = |x_1| + |x_2| = \|x\|_1 \quad \text{for all } x \in \mathbb{R}^2.$$

*Consequently  $T$  is an isometry from  $\ell_1^2$  onto  $\ell_\infty^2$ .*

*Proof.* The displayed identity is Lemma 2 with  $a = x_1, b = x_2$ . In particular  $T$  preserves norms, hence is injective; and  $T$  is surjective with inverse

$$T^{-1}(y_1, y_2) = \left( \frac{y_1+y_2}{2}, \frac{y_1-y_2}{2} \right),$$

so  $T$  is a linear isometry of  $\ell_1^2$  onto  $\ell_\infty^2$ .  $\square$

Geometrically,  $T$  maps the four vertices of the  $\ell_1^2$  unit ball to the four corners of the  $\ell_\infty^2$  unit ball:

$$(1, 0) \mapsto (1, 1), \quad (-1, 0) \mapsto (-1, -1), \quad (0, 1) \mapsto (1, -1), \quad (0, -1) \mapsto (-1, 1).$$

Since  $T$  is linear and the diamond is the convex hull of its vertices,  $T$  carries the diamond onto the square.

### 3 The counterexample at $n = 2$

**Theorem 4.**  $d(\ell_1^2, \ell_\infty^2) = 1$ . In particular  $d(\ell_1^2, \ell_\infty^2) \neq \sqrt{2}$ .

*Proof.* By Proposition 3,  $T$  is a bijective linear isometry from  $\ell_1^2$  onto  $\ell_\infty^2$ , and its inverse  $T^{-1}$  is a bijective linear isometry from  $\ell_\infty^2$  onto  $\ell_1^2$ . Hence, for the operator norms defining the distance,

$$\|T\|_{\ell_1^2 \rightarrow \ell_\infty^2} = \sup_{x \neq 0} \frac{\|Tx\|_\infty}{\|x\|_1} = 1, \quad \|T^{-1}\|_{\ell_\infty^2 \rightarrow \ell_1^2} = \sup_{y \neq 0} \frac{\|T^{-1}y\|_1}{\|y\|_\infty} = 1.$$

Therefore  $d(\ell_1^2, \ell_\infty^2) \leq \|T\| \|T^{-1}\| = 1$ . Since  $d(E, F) \geq 1$  always (as  $T^{-1}T = \text{id}$  forces  $1 = \|\text{id}\| \leq \|T\| \|T^{-1}\|$ ), we conclude  $d(\ell_1^2, \ell_\infty^2) = 1$ . Finally  $1^2 = 1 \neq 2 = (\sqrt{2})^2$  and both numbers are positive, so  $1 \neq \sqrt{2} \approx 1.41421$ .  $\square$

**Corollary 5.** *The sentence “the distance between  $\ell_1^n$  and  $\ell_\infty^n$  is exactly  $\sqrt{n}$ ” is false. It fails at  $n = 2$ , where the true distance is 1.*

*Proof.* Immediate from Theorem 4: the asserted value for  $n = 2$  is  $\sqrt{2}$ , but the actual distance is 1.  $\square$

*Remark 6* (Why this is the wrong quantity, not a rounding error). It is a standard fact that

$$d(\ell_1^n, \ell_2^n) = \sqrt{n}, \quad d(\ell_2^n, \ell_\infty^n) = \sqrt{n},$$

the “Goldstine ratio”. This is the source of the  $\sqrt{n}$  appearing in the conjecture. However, the pair  $(\ell_1^n, \ell_\infty^n)$  is different: since  $(\ell_1^n)^* = \ell_\infty^n$ , duality gives  $d(\ell_1^n, \ell_\infty^n) = d(\ell_\infty^n, \ell_1^n)$ , and the identity map already shows  $d(\ell_1^n, \ell_\infty^n) \leq n$ . More importantly, the distance between  $\ell_1^n$  and  $\ell_\infty^n$  is of order  $n$ , not of order  $\sqrt{n}$ : there are absolute constants  $c, C > 0$  with

$$cn \leq d(\ell_1^n, \ell_\infty^n) \leq Cn.$$

This is the endpoint  $p = 1, q = \infty$  of the classical two-sided estimate  $d(\ell_p^n, \ell_q^n) \asymp n^{1/p-1/q}$  for  $1 \leq p \leq q \leq \infty$ ; here  $1/p - 1/q = 1$ . Thus  $\sqrt{n}$  is asymptotically the wrong order of magnitude, not an off-by-a-small-constant error, and replacing it by any multiple of  $\sqrt{n}$  cannot repair the statement. The decomposition  $d(\ell_1^n, \ell_\infty^n) \leq d(\ell_1^n, \ell_2^n) d(\ell_2^n, \ell_\infty^n) = n$  and the matching order- $n$  lower bound are the correct context.

*Remark 7* (On the asymptotic reading). The conjecture’s surrounding discussion concerns the asymptotic diameter  $d_n$  of  $n$ -dimensional Banach spaces and the limit  $d_n/n \rightarrow 1$ . The specific sentence we refute, however, is stated with “exactly  $\sqrt{n}$ ” and carries no “asymptotically” / “as  $n \rightarrow \infty$ ” qualifier. Reading it as an exact identity for every  $n$  makes it false at  $n = 2$ ; reading it as merely asymptotic is not what the text says, and even under that reading  $\sqrt{n}$  is the wrong order since the true distance grows like  $n$ . We therefore mark the  $\ell_1^n$ – $\ell_\infty^n$  sub-claim of conjecture 00000000938 as **false**, independently of the conjecture’s remaining claims about  $d_n$ .

## 4 Formal verification

The identity of Lemma 2, the isometry property of  $T$ , the vertex correspondence, and the arithmetic fact  $1^2 \neq 2$  are formalised in core Lean 4 (toolchain `leanprover/lean4:v4.33.1`, only `import Std`, no `Mathlib`, no `sorry`). See `lean4/`. A standalone numerical reproduction over all integer vectors in  $[-25, 25]^2$  is provided in `reproduce.py`.